DG^2-TCR: An Adaptive Clouds Removal Network for Optical Remote Sensing Images Using SAR-Driven Dual-Flow Fusion Guidance

Xianjun Gao, *Member, IEEE*, Jinhui Yang, Xudong Xie, Yuanwei Yang, Nan Wang, Xinran Cao, Bin Du, Meilin Tan, Lei Xu, and Yuan Kou

***Abstract*— Clouds in optical remote sensing images significantly limit image utilization. Traditional cloud removal methods using single or multi-temporal data sources struggle to ensure reliable reconstruction for thick cloud (TKC) areas. Synthetic aperture radar (SAR) images are increasingly used to recover information obscured by clouds, but their performance in cloud-obscured regions is unstable. Therefore, an adaptive cloud removal network for remote sensing images, named DG^2-TCR, is proposed based on SAR-driven dual-flow fusion guidance (DFG). DG^2-TCR uses SAR and optical remote sensing imagery (ORSI) to construct DFG, including local spatial-spectral feature reconstruction (LSSFR) flow and global texture feature compensation (GTFC). LSSFR, driven by ORSI and SAR, efficiently extracts useful features in noncloud areas and focuses on local information reconstruction using the designed spatial-spectral features inference reconstruction block (SSIRB). Based on SAR images, GTFC guides the compensation of global texture information. DFG can adaptively extract features and reconstruct missing information from local and global scales. The public SEN12MS-CR-TS dataset is divided into four sub-datasets with different coverage to evaluate the recovering capability in varying clouds. Experiments show that the peak signal-to-noise ratio (PSNR), structural similarity index (SSIM), root mean square error (RMSE), Fréchet inception distance (FID), and normalized cross correlation (NCC) indicator values on four sub-datasets and the SIMLE-CR datasets are better than the seven comparison methods. Furthermore, the ablation experiments show that the generalization and robustness of this proposed method on images with different cloud coverage are better than other comparison methods. Therefore, DG^2-TCR can reliably recover information on cloud occlusions with various coverage and thickness, which is significant for cloud removal in practical applications.**

***Index Terms*— Cloud removal, dual-flow fusion guidance (DFG), optical remote sensing image, spatial-spectral inference and reconstruction, synthetic aperture radar (SAR).**

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Xianjun Gao, Jinhui Yang, Xudong Xie, Yuanwei Yang, and Xinran Cao are with the School of Geosciences, Yangtze University, Wuhan 430100, China (e-mail: junxgao@yangtzeu.edu.cn; 2022710467@yangtzeu.edu.cn; 2024720538@yangtzeu.edu.cn; yyw_08@yangtzeu.edu.cn; 2023720560@yangtzeu.edu.cn).

Nan Wang is with the School of Information and Electronics, Beijing Institute of Technology, Beijing 10081, China (e-mail: wang_nan@bit.edu.cn).

Bin Du and Meilin Tan are with the Region Surveying and Mapping Geographic Information Center, Hohhot 010050, China (e-mail: 596482264@qq.com; tanmeilin@whu.edu.cn).

Lei Xu is with China Railway Design Corporation, Tianjin 300308, China (e-mail: xulei2013@whu.edu.cn).

Yuan Kou is with the First Surveying and Mapping Institute of Hunan Province, Changsha, Hunan 421001, China (e-mail: kouyuan@xcyy.net.cn).

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NOMENCLATURE

ORSI	Optical remote sensing imagery.
SAR	Synthetic aperture radar.
DFG	Dual-flow fusion guidance.
LSSFR	Local spatial-spectral feature reconstruction.
GTFC	Global texture feature compensation.
SSIRB	Spatial-spectral features inference reconstruction block.
DAGC	Dual-activation gated convolutional block.
CNNs	Convolutional neural networks.
CAB	Channel attention block.
TNC	Thin cloud.
SPC	Small patchy cloud.
LPC	Large patchy cloud.
TKC	Thick cloud.
CMLP	Contextual multiple linear prediction.
NSPI	Neighborhood similar pixel interpolator.
cGAN	Conditional generative adversarial network.
U-SSIRB	U-shape SSIRB.
GTE	Global feature encapsulation.
GTC	Global texture compensation.
FE	Feature extraction.
FR	Feature reconstruction.
GC	Gated convolution.
SE	Squeeze-and-excitation.
PSNR	Peak signal noise ratio.
mse	Mean squared error.
RMSE	Root mean square error.
SSIM	Structural similarity index.
SAM	Spectral angle mapper.
NCC	Normalized cross correlation.
FID	Fréchet inception distance.

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I. Introduction

OPTICAL remote sensing imagery (ORSI) is a vital data source for numerous Earth observation missions, including routine monitoring of surface resources, disaster assessment, and so on. However, due to atmospheric influences, an average of 67% of the Earth's surface is covered by clouds, making cloud cover at various scales almost inevitable in ORSI [1], [2], [3], [4], [5], [6]. Consequently, the ground object information is obscured and lost, reducing the temporal and spatial information availability of surface observation images [7], [8]. Cloud removal aims to recover the information obscured by clouds, which are useful for improving image availability and enhancing continuous observation in time series analysis. Therefore, cloud removal in ORSI significantly enhances the image quality and utilization value.

However, the image information loss shaded by clouds varies with the thickness of the cloud cover. As shown in Fig. 1, the common TNC and TKCs caused different shading influences [9], [10]. Generally, TNC and TKC occlusion adopt different principles for cloud removal. Although TNCs will block the ground objects, some detailed information under the clouds will still be retained [11], [12]. Therefore, TNCs are viewed as ordinary noise in TNC removal, such as Gaussian and pretzel in the image. Then, TNCs can be effectively removed using filtering methods [12], [13], [14], [15] such as Gaussian low-pass filtering, homomorphic filtering, median filtering, or modeling methods such as Markov random fields. In contrast, TKCs will cause severe loss of information, making it hard to restore the factual obscured ground object information in the same image effectively. Moreover, TKCs occur more widely in time and range. Therefore, effectively removing TKCs from images is of great research significance for improving the utilization of remote-sensing images [16].

Existing methods for TKC removal from remote sensing images can be roughly categorized into three categories [17], [18], [19], [20]: single-source-based, multitemporal-based, and multisource-based. Single-source-based TKC removal methods infer or reconstruct the area obscured by TKCs based on the information of the cloud-free area in the same image. The cloud-free areas should be detected first to provide the spectral and spatial reference information. On the basis of the feature similarity between cloud-cover and cloud-free regions, spatial interpolation methods, texture synthesis, and image inpainting principles are used to reconstruct the covered information [19], [20], [21], [22]. These methods effectively process remote sensing images with low resolution and few clouds. However, the performance degrades significantly as the image resolution and cloud cover increase. The multitemporal-based cloud removal method uses interpolation and splicing of two or more cloud-free remote sensing images captured by the sensors over the same area at different time to replace pixels in cloud-covered regions [23], [24], [25]. These methods fuse multitemporal images, enhancing the reliability of the final cloud-free output [26]. They are suitable for areas where the surface changes slowly and are particularly effective in agriculture and forest monitoring applications. The time series information can be used to improve the spatial and temporal resolution of images and ensure the integrity of observation data. However, such methods are limited by the revisit period of the sensors and the seasonal variations. If the satellites revisit the same area for an extended period, the acquired images do not guarantee that the ground objects will be compensated promptly. Moreover, in regions with prolonged cloud cover and seasonal variations (e.g., southern China during the rainy season, the Amazon rainforest), securing sufficient high-quality, cloud-free images is challenging. Multisource-based TKC removal methods leverage data from various types of sensors to address the limitations of spatial and multitemporal cloud removal approaches, which often rely too heavily on imagery from similar sensors [27]. The quality of spatial or temporal information from a single-image source greatly impacts the effectiveness of cloud removal [28]. Compared with optical imaging technology, SAR can directly capture ground texture through the cloud cover. As a result, many researchers use SAR images to aid in cloud removal from ORSI. With the advancement of deep learning in remote sensing, CNNs have been utilized to fuse SAR data and optical data, achieving effective cloud compensation in remote sensing imagery [28], [29], [30], [31], [32]. These methods enhance the efficiency and effectiveness of cloud removal in remote sensing images and improve the utilization of multisource data [32], [33]. Although these methods can remove clouds from remote sensing images, their application scenarios are limited. Current methods perform well in cases of TNCs and minimal cloud cover. However, as the cloud thickness and coverage increase, the amount of lost information in remote sensing images also increases, significantly reducing the effectiveness of existing cloud removal methods. Overall, cloud removal in ORSI still faces the following challenges.

1) Adaptive cloud reconstruction with varying cover conditions is needed to be improved. Cloud cover with different thicknesses leads to significant differences in the loss of surface information. TNC and TKC removal adopted different principles. Existing methods are either designed specifically for removing thin or fragmented clouds or for TKCs, and there is a lack of effective techniques that can adaptively remove clouds based on the degree of obstruction.
2) The TKC-cover reconstruction accuracy is a challenge. Traditional methods supplement the missing surface details using other temporal images, resulting in FR discrepancies. Integrating SAR data for TKC removal helps mitigate errors. However, the inherent heterogeneity between SAR and optical images makes alignment and matching across spatial, spectral, and geometric dimensions challenging. Furthermore, the complex and varied characteristics of surface features complicate feature mapping, adding to the uncertainties in reconstructing cloud-covered information.

Therefore, we propose an adaptive cloud removal network DG^2-TCR for ORSI cloud removal. The network employs a DFG structure to separate the cloud removal task into two subtasks: LSSFR and GTFC. LSSFR leverages the U-shaped structure of the SSIRB (SSIRB) to extract valid information

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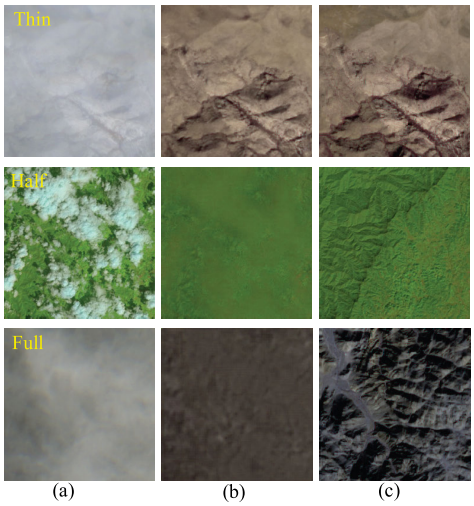


Fig. 1. Comparison of different cloud occlusion and cloud-removal effects. (a) Clouds images with different coverage. (b) Cloud-removal images using nonlinear activation free network (NAFNet) [34]. (c) Cloud-free images. The cloud-removal results get worse with the increase in cloud thickness and coverage.

from noncloud areas, suppress invalid information in cloud-obscured regions, and infer missing details within the feature layer for reconstruction. GTFC utilizes priori knowledge from SAR to refine and correct texture details within the reconstructed feature layer. The two tasks converge at the end of the network to adaptively generate a cloud-free ORSI.

The main contributions of this article are outlined as follows.

1) This article proposes an adaptive cloud removal network, DG^2-TCR, that uses SAR to remove clouds in ORSI. DG^2-TCR can adaptively complete the reconstruction of missing information under the clouds according to different cloud thicknesses, shapes, and coverage rates.
2) A DFG structure is designed to mine SAR fully to guide the reconstruction of lost information from global and local perspectives. DFG concludes with LSSFR and GTFC. These subtasks restore missing data in the cloud-covered areas and refine global texture features in the reconstructed regions.
3) An SSIRB module is explored to implicitly distinguish between cloud and noncloud pixels and infer missing information for reconstruction. SSIRB integrates the dual-gated convolutional block and CAB, which can effectively extract the feature layer information and reconstruct missing details across spatial and channel dimensions. SSIRB can implicitly complete the reasoning and reconstruction of different clouds.
4) A paired dataset of SAR and optical images for the cloud removal task has been reorganized. To validate the adaptive cloud removal ability of DG^2-TCR, the SEN12MS-CR-TS dataset is divided into four sub-datasets, including TNC, SPC, LPC, and TKC. The results show that DG^2-TCR performs best in the sub-datasets.

The rest of the article is organized as follows. The related researches are discussed in Section II. The details of the DG^2-TCR are presented in Section III. Section IV describes the experimental analysis. The ablation and robustness exploration of the network are discussed in Section V. We summarize the work of this article in Section VI.

II. Related Studies

This section discusses three categories of cloud removal methods, including single-source-based, multisource-based, and multitemporal-based methods.

A. Single-Source-Based Cloud Removal Methods

Single-source-based cloud removal methods remove clouds without relying on additional images. These approaches primarily use interpolation or image-denoising algorithms to eliminate clouds from remote sensing images. Spatial interpolation utilizes the spectral information from the surrounding cloud-free regions to perform interpolation calculations, replacing the missing information with the interpolated results. For instance, the Kriging interpolation method was adapted and applied to the cloud removal task for better results [21]. While the overall cloud removal performance is acceptable, the interpolation areas still exhibit slight differences compared to cloud-free regions. Lee et al. [35] proposed a method for Landsat7 satellite to restore cloudy images using a 1-D interpolation and filtering techniques called gap interpolation and filtering. However, since the interpolation algorithm operates within cloud-covered regions, it first requires accurate detection of the cloud-specific location and coverage within the image. The effectiveness of such methods declines significantly as image resolution and cloud cover increase.

B. Multitemporal-Based Cloud Removal Methods

Multitemporal-based cloud removal methods primarily utilized cloud-free images captured in the same area at different times. Lin et al. [18] and Kalkan and Maktav [36] employed an information cloning method to extract the supplementary information from multitemporal images for reconstructing cloud-obscured regions. Melgani [37] utilized temporal image data from the Landsat7 ETM+ sensor to reconstruct cloud-covered regions. The CMLP method was applied to recover cloud-covered areas within a multitemporal, multispectral image sequence by matching simultaneous image data and detecting cloud cover. Then, Zhu et al. [38] proposed a TKC removal method based on the improved NSPI. However, CMLP struggles to remove clouds effectively when the cloud coverage is extensive or no cloud-free auxiliary images are available. The performance of both CMLP and NSPI is constrained by the extent of cloud cover in the image and the availability of auxiliary images. Meng et al. [39] proposed the closest spectral fit method for cloud removal, which shows strong effectiveness on Landsat and QuickBird series, unaffected by cloud size, thickness, or density. However, this method still depends on the availability of cloud-free auxiliary images, which remain a limitation.

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In recent years, deep CNNs have achieved notable success in remote sensing. Zhang et al. [40] applied CNNs to extract multiscale features from nontemporal images, facilitating the restoration of missing data in cloud-covered regions of target images. This method demonstrates effective cloud removal across various scenes and temporal phases in remote sensing images captured by the same sensor. Tu et al. [41] utilize a fully connected tensor network to decompose the global correlations in multitemporal images and apply weighted group sparsity to characterize cloud features, resulting in effective removal. These methods require short imaging time intervals between the cloudy image and the cloud-free reference image and assume minimal changes in features to ensure optimal cloud removal results. Otherwise, multitemporal phase cloud removal methods may not guarantee the validity and reliability of the compensated feature information in cloud-covered areas. Additionally, this method relies on highly accurate cloud detection results, restricting its applicability range.

C. Multisource-Based Cloud Removal Methods

The multisource remote sensing imagery leverages data from different sensors to remove clouds. Some researchers [42], [43], [44] used high-temporal, low-spatial resolution images from various optical sensors as complementary data to fill the cloud-covered area. However, the spatial resolution is often too low to reconstruct high-resolution images effectively [45]. In addition, all optical bands are affected to varying degrees by cloud cover, making them insufficient for fully compensating data loss in dense cloud-obscured regions; therefore, relying solely on ORSI cannot effectively recover information in cloud-covered areas. In contrast, SAR imaging technology can capture images in most weather conditions [46], [47], [48], as it can penetrate clouds to obtain precise surface features. Thus, SAR imagery can enhance the reliability of reconstructed information when used alongside ORSI for cloud removal. Meraner et al. [30] employed a deep residual neural network architecture and designed a cloud-adaptive loss function to integrate SAR images and optical remote sensing images, using these combined features to remove clouds from Sentinel-2 imagery. However, they overlooked potential information loss. Grohnfeldt et al. [31] proposed a cGAN architecture that combines SAR and optical multispectral images for cloud removal. Their approaches utilized cloud-contaminated ORSI and SAR-assisted images to generate clear, cloud-free, and fog-free ORSI. Wang et al. [49] proposed a novel unified spatial-spectral residual network that uses SAR as auxiliary reference data to facilitate cloud removal in ORSI. The network divides the information reconstruction task into FE, fusion, and reconstruction modules. It employs GC and a null spectral attention mechanism to differentiate between cloud pixels and cloud-free pixels, enabling the effective reconstruction of global spatial-spectral information.

HS^{2}P [50] introduces a hierarchical spectral and structural fusion network to mitigate noise interference in SAR and ORSI data fusion. Its deep architecture with stacked residual groups adaptively selects relevant features, reducing noise and enabling cloud and shadow removal in ORSI. TransGAN-CFR Li et al. [51] proposes a transformer-based GAN with a hierarchical encoder–decoder structure and multihead self-attention, enhanced by depthwise convolutions and gating mechanisms, to reconstruct cloud-free multispectral images from SAR and optical inputs. CR-Former [52] leverages a cloud simulation strategy to create the SMILE-CR dataset for Landsat-8 and employs a Transformer-based network with multihead dense and sparse attention alongside a multiscale-gated feedforward network to remove cloud efficiently.

SAR has become a crucial data source for reconstructing missing information in ORSI due to its unique cloud-penetrating capabilities and high reliability. While existing methods leverage SAR as prior knowledge, they often assist in removing TKCs from ORSI. They often overlook the potential of SAR images to provide global guidance. Consequently, although these methods can remove clouds, they struggle with scenes containing TKCs and extensive coverage. Thus, this article proposes a novel network for effective cloud removal in ORSI.

III. Methodology

In this section, the principle of DG^2-TCR is illustrated. First, an overview of DG^2-TCR is presented. This is followed by an explanation of the dual-flow guidance idea. Next, the detailed workings of LSSFR and GTFC are described. The section then continues with a description of the final loss function. Finally, the complete workflow is outlined.

A. Overview of the DG^2-TCR Network

As illustrated in Fig. 2, DG^2-TCR incorporates SAR images as prior knowledge to enhance the reconstruction of cloud-covered optical images. DFG uses SAR to reconstruct missing information from LSSFR and GTFC subtasks. LSSFR-fused SAR and ORSI to put in a U-SSIRB, based on dual-activation gated convolutional block (DAGC) and CAB, To extract essential features and reconstruct missing features. GTFC passes through only one convolution and activation layer to minimize global feature loss. Thus, global texture features can be captured. Then, LSSFR features are introduced to supervise and guide the compensation of the local texture features that have not been effectively recovered from the global scale. Finally, the highly nonlinear SAR-to-optical feature transformation is learned and implicitly performed in the network to infer the reconstruction information from cloud to cloud-free images. A balanced reconstruction of results is achieved between local details and global consistency.

B. DFG Structure of DG^2-TCR

As shown in Fig. 2, to enhance the recovery of detailed information, DG^2-TCR introduces SAR images to construct LSSFR and GTFC. The information flows are delivered and fused through LSSFR and GTFC, respectively.

Denote $I_{\text{SAR}} \in \mathbb{R}^{f_s \times H \times W}$ and $I_{\text{cloud}} \in \mathbb{R}^{f_o \times H \times W}$ are the SAR and ORSI, respectively, f_s and f_o are the number of feature maps, and H and W represent the height and width. LSSFR

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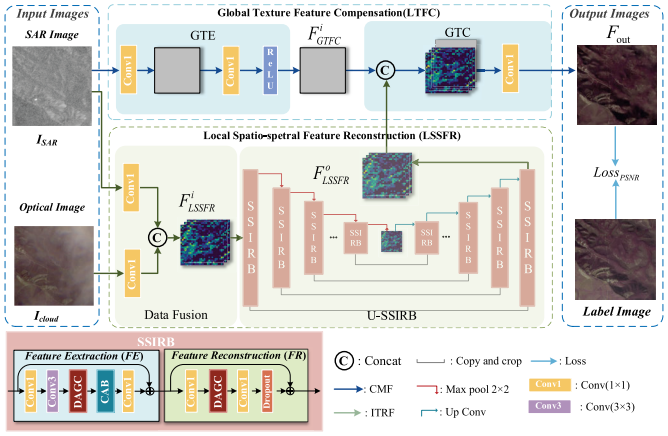


Fig. 2. Structure of the DG^2-TCR model.

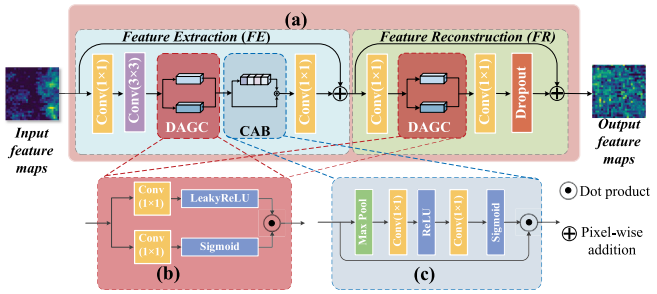


Fig. 3. FR block structure. (a) SSIRB. (b) DAGC. (c) CAB.

task conducts multimodal feature fusion and local FR on the I_{SAR} and I_{cloud}. They are concatenated and fused through f_u, defined as (1). Then, LSSFR reconstructs the local missing information of the cloud-covered feature layer through a U-SSIRB, defined in (2)

$$F^i_{\mathrm{LSSFR}} = f_u(I_{\mathrm{SAR}}, I_{\mathrm{cloud}}) \tag{1}$$

$$F^o_{\mathrm{LSSFR}} = f_{U-SSIRB}\left(F^i_{\mathrm{LSSFR}}\right) \tag{2}$$

where f_u represents the data fusion function. F^i_{LSSFR} is the encapsulated local information reconstruction. F^o_{LSSFR} is the reconstructed feature layer output of LSSFR. $f_{U-SSIRB}$ is the calculation process of the U-shaped feature reasoning reconstruction structure built by SSIRB.

Second, the global texture feature of I_{SAR} will be encapsulated and used for compensation through GTFC. F^i_{GTFC} is obtained through the GTE function f_{GTE} according to (3). Then, F^i_{GTFC} will be fused with F^o_{LSSFR} by the GTC function f_{GTC}, defined in (4) to obtain the output F_{out}

$$F^i_{\mathrm{GTFC}} = f_{\mathrm{GTE}}(I_{\mathrm{SAR}}) \tag{3}$$

$$F_{\mathrm{out}} = f_{\mathrm{GTC}}\left(F^o_{\mathrm{LSSFR}}, F^i_{\mathrm{GTFC}}\right) \tag{4}$$

where F^i_{GTFC} is obtained through convolution and activation. f_{GTE} and f_{GTC} are the GTE and GTC functions, respectively. F_{out} is the final output of the model.

Thus, DFG guided the dual-flows comprising of LSSFR and GTFC to reconstruct the missing information.

C. Local Spatial-Spectral Features Reconstruction

LSSFR is constructed with UU-SSIRB. As shown in Fig. 3, a basic unit SSIRB consists of FE and FR. First, in the FE, DAGC and a CAB are designed to extract valuable feature information from both spatial and channel dimensions. The initial feature layer is fused with the extracted feature layer through a skip connection to prevent the loss of global feature information and effectively integrate local features. Then, in the FR, only one DAGC is used since the null spectrum features have been fully extracted earlier. In addition, a skip connection is used to ensure seamless fusion between reconstructed features in cloud and cloud-free regions. Each SSIRB employs convolutional layers with varying kernels and step sizes to adapt feature layer shapes and perform upsampling and downsampling efficiently.

1) Dual Activation Gated Convolutional Block: Since the cloud removal task has many invalid pixels in the cloud-covered area, the ordinary convolution cannot handle them because they will gradually become valid pixels in the network, leading to color differences and blurring of the reconstructed image [53]. Therefore, it is necessary to distinguish valid pixels from invalid pixels. Thus, we introduced GC to make up for the defects of ordinary convolution.

To further enhance cloud/cloud-free discrimination and FE capabilities, we proposed a DAGC block that improved upon traditional GC. DAGC provides learnable dynamic features for each channel at the spatial position of each layer. Specifically, the Sigmoid and Leaky ReLU [54], [55] activation functions are introduced into the two branches of GC, as shown in Fig. 3(b). The sigmoid function normalizes the output of each neuron to [0, 1] for extracting feature layer information conductively. In addition, the sigmoid function has a smooth gradient, which can effectively avoid unreasonable output values. The leakyReLU function, improved based on the ReLU function, inherits the advantage of computational speed and alleviates the problem of the neuron parameters not being updated by retaining the negative value. Therefore, leakyReLU helps improve the stability and training efficiency of the model in different scenarios. Specifically, it can be described as follows:

$$\mathrm{Gatting}(y, x) = \sum\sum W_g \times F + b_g \tag{5}$$

$$F'(y, x) = \Phi(\mathrm{Gatting}(y, x)) \odot \sigma(\mathrm{Gatting}(y, x)) \tag{6}$$

where W_g is the weight of the convolution operation, and b_g is the bias parameter. F is the feature layer of the input DAGC. Φ is the leakyReLU activation function, σ is the Sigmoid activation function. $\Phi(\mathrm{Gatting}(\cdot))$ and $\sigma(\mathrm{Gatting}(\cdot))$ are GCs. $F'(y, x)$ is the output feature after DAGC processing.

In sum, DAGC can enhance the network by learning the distinction between valid and invalid values and helping the network entirely use the information in the valid value area to reconstruct cloud-free images.

2) Channel Attention Block: Although DAGC can effectively extract feature information at different spatial scales, its ability to extract interchannel features is insufficient. Therefore, we introduce CAB to obtain global channel information for spatial and spectral correlation during reconstruction.

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As shown in Fig. 3(c), the channel attention model, namely the SE block, is adopted. SE is divided into compression and activation to compress the global spatial information and assign different weights to each channel. The input feature vector $X \in \mathbb{R}^{H\times W\times C}$ is first reshaped by global average pooling to obtain a new feature vector $X_C \in \mathbb{R}^{1\times 1\times C}$ in the compression part. Then, X_C is processed by a multilayer perceptron f_{MLP} in the activation to predict the feature vectors $E_C \in \mathbb{R}^{1\times 1\times C}$ of each channel, which is defined as follows:

$$E_C = f_{\text{MLP}}(X_C). \tag{7}$$

To accurately apply the obtained channel feature vector E_C to the original element feature vector X, dimension mapping is performed on to get $E'_C \in \mathbb{R}^{H\times W\times C}$. Finally, it is used as a channel feature to act on the initial element feature layer. The final result E is calculated as follows:

$$X_{\text{out}} = X_C \odot E'_C. \tag{8}$$

D. Global Texture Feature Compensation

A GTFC is designed to reconstruct the information with a TKC cover. In this task, the SAR image is input again to guide the output results of LSSFR. As shown in Fig. 2, GTFC is designed with two inputs, including I_{SAR} and F^o_{LSSFR}. Both input ends use 1×1 convolution blocks, which adjust the shape of the input feature layer and extract and enhance nonlinear features. This design improves the model's expressive power and reduces the likelihood of gradient vanishing. In addition to the convolution block, the SAR image input end incorporates a ReLU activation function. This activation enhances the model's nonlinear characteristics and improves its ability to learn and utilize SAR image features in complex scenes. To minimize the loss of SAR image texture details and enhance GTFC, the SAR image encapsulation uses just one convolution layer and one ReLU activation layer. Thus, I_{SAR} is processed through a convolution layer and an activation layer, encapsulating global texture features F^i_{GTFC}, defined in the following equation:

$$F^i_{\text{GTFC}} = f_{\text{GTE}}(I_{\text{SAR}}) = S_{\text{ReLU}}((I_{\text{SAR}} * k) + b) \tag{9}$$

where S_{ReLU} is ReLU activation function, $*$ represents convolution operation, k is the convolution kernel size, b is the bias, and F^i_{GTFC} is the encapsulated global texture feature layer.

After convolutional feature alignment, F^i_{GTFC} is concatenated with F^o_{LSSFR}. This combined feature undergoes the same convolution process described earlier, producing the final output F_{out}, as shown in the following equation:

$$F_{\text{out}} = f_c\left(F^i_{\text{GTFC}}, F^o_{\text{LSSFR}}\right) * k + b. \tag{10}$$

This process uses SAR data to supplement the global texture details lost in the local reconstruction feature layer. The spliced feature layer undergoes convolutional sampling with varying dimensions, enabling cross-channel interaction between the SAR and local reconstruction features. Thus, the resolution of missing information in cloud-occluded areas is enhanced, and image clarity is improved.

E. Loss Function

A PSNR is explored as the loss function to better align with the specific challenges of TKC removal. First, the three channels of the predicted result I_{out} and the ground truth image I_{gt} are normalized and converted into grayscale channel feature maps I_{out} and I_{gt}. Since the human eye is more sensitive to changes in grayscale than to changes in three-band colors, the grayscale channel can precisely reflect the changing characteristics of each pixel:

$$f_{(I_R,I_G,I_B)} = C_1 I_R + C_2 I_G + C_3 I_B \tag{11}$$

where I_R, I_G, and I_B are the red, green, and blue feature information of the images (I_{out} or I_{gt}), respectively. Generally, $C_1 = 0.299$, $C_2 = 0.587$, $C_3 = 0.114$.

Next, the mse can be calculated between I_{out} and I_{gt}.

$$\text{mse} = \frac{1}{H \times W}\sum_{i=1}^{H}\sum_{j=1}^{W}(I_{\text{out}} - I_{\text{gt}})^2 \tag{12}$$

where H and W are the height and width of the feature map. $H \times W$ represents the number of pixels in the feature map. The PSNR is calculated based on the mean square error, and the calculation equation is as follows:

$$\text{Loss}_{\text{PSNR}} = \frac{1}{B}\sum_{b=1}^{B}\left[10\log_{10}\left(\frac{(2^n-1)^2}{\text{mse}+\varepsilon}\right)\right] \tag{13}$$

where B is the batch size during training, and $\text{Loss}_{\text{PSNR}}$ represents the average loss in each batch. In order to ensure that zero values are not encountered when calculating the logarithm, a smoothing term ε is added to the mse, and the value is 1×10^{-8} during the calculation process. The smoothing term prevents the denominator from zero and ensures numerical stability with minimal impact on the final result.

F. Workflow

The overall workflow of DG^2-TCR is shown in Algorithm 1. First, the SAR image I^i_{SAR} and the cloud ORSI image I^i_{cloud} are fused into the feature I^i_{in}; then, I^i_{in} is reconstructed through the U-shaped structure inference composed of SSIRB to obtain the primary reconstruction feature F^i_{out}. Next, the SAR image I^i_{SAR} corrects the texture details of the primary reconstruction feature F^i_{out}. Finally, the cloud-free image I^i_{out} is obtained after network inference and reconstruction. The cloud-removal image I^i_{out} and the cloud-free image I^i_{nocloud} are computed with loss and back-propagated into the network for gradient updating. After sub-iterations, the training process of the DG^2-TCR network is finalized.

Since DG^2-TCR is based on CNN, its complexity can be illustrated by Time $\sim O(M^2 \cdot K^2 \cdot C_{\text{in}} \cdot C_{\text{out}})$. M is the side length of the output feature map from the convolutional kernel, and K represents the side length of the convolutional kernel. C_{in} and C_{out} represent the channel number of input and output feature maps. Since the overall structure of DG^2-TCR is fixed, both C_{in} and C_{out} are fixed values. The dimensions and number of channels for both the input and output of the network are also fixed. Therefore, the complexity of the proposed method

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Algorithm 1 Train Process for the DG^2-TCR

Input: SAR image $\{I^i_{SAR}\}^N_{i=1}$, cloud ORSI $\{I^i_{cloud}\}^N_{i=1}$, maximum iteration T

Output: The network weights θ_D

$t \leftarrow 0$ and randomly initialize θ_D

while $t \leftarrow 0$ and NOT Converged **do**

Randomly sample $[N]$ to obtain Ω with $|\Omega| = m$;

for $i \in \Omega$ **do**

Data fusion: Synthesize F^i_{LSSFR} using I^i_{SAR} and I^i_{cloud} according to $F^i_{LSSFR} = f_u(I_{SAR}, I_{cloud})$

USSIRB: Synthesize F^o_{LSSFR} using F^i_{LSSFR} and according to $F^o_{LSSFR} = f_{U-SSIRB}(F^i_{LSSFR})$

GTE: Synthesize F^i_{GTFC} using and according to $F^i_{GTFC} = f_{GTE}(I^i_{\mathrm{SAR}})$

GTC: Synthesize using F_{out} and according to $F_{out} = f_{GTC}\left(F^o_{LSSFR}, F^i_{GTFC}\right)$

DG^2-TCR uses inference to obtain;

end for

Update θ_D by $\nabla\theta_D(Loss_{PSNR})$

$t \leftarrow t + 1$

end while

return θ_D

is fixed and will not increase or decrease due to external factors such as changes in the number of channels.

IV. Experiments Results and Analysis

A. Experimental Data

The training dataset required should ideally encompass various land cover types, such as forests, cities, oceans, deserts; therefore, the SEN12MS-CR-TS [56] and SMILE-CR [52] are the freely available public dataset for cloud removal. SEN12MS-CR-TS, is a multimodal that contains Sentinel-1 and Sentinel-2A and a multitemporal dataset for cloud removal. The average time interval of SEN12MS-CR-TS is 2.61 days. Multitemporal data make the cloud-free image a more reliable ground truth. It can be accessed at https://patricktum.github.io/cloud_removal/sen12mscrts/. SMILE-CR is a global and multimodal dataset, including Landsat8, Sentinel-1, and MODIS data, which link is https://www.kaggle.com/datasets /yuxiawhu/smile-cr

The SEN12MS-CR-TS and SIMLE-CR offer data segmented into 256 × 256 patches. The dataset patches are preprocessed. First, we extracted the RGB bands from Sentinel-2A to construct ORSI. Then, the SAR image is normalized to [0, 255], facilitating the integration of SAR and ORSI data. Third, to test our methods on different clouds, we deliberately select a representative scene in the SEN12MS-CR-TS dataset to construct a sub-dataset, including five categories: cloud-free, TNCs, SPCs, LPCs, and TKCs. They are divided according to three crucial factors: the cloud coverage ratio, the extent of coverage, and the morphology of the clouds. We employed the MTDR-Net [57] to detect clouds in each reconstructed image, conducting a visual analysis to assess the cloud morphology coverage. The cloud-free images are saved as labeled data. The remaining images are categorized into distinct sub-datasets: TNC scene, SPC scene, LPC scene, and TKC scene. The cloud coverage of the sub-dataset is shown in Table I. Fig. 4 displays an example of classifying one location scene. To streamline the training process, the images are cropped into patches of 256 × 256 pixels, and Table II summarizes the number of patches allocated for training and testing purposes.

TABLE I

Proportion of Cloud-Covered Area to the Whole Image in Different Scenes

Sub-dataset Type	Classification	r: Cloud proportion rate
TNC	thin cloud scene	$r < 10\%$
SPC	small patchy cloud scene	$10\% \le r < 30\%$
LPC	large patchy cloud scene	$30\% \le r < 60\%$
TKC	thick cloud scene	$r \ge 60\%$

TABLE II

Number of Patches for Different Sub-Datasets

Dataset	Training sample	Validation sample	Testing sample
TNC	3880	485	484
SPC	6175	772	771
LPC	4439	692	692
TKC	5944	743	742
SMILE-CR	1000	200	200

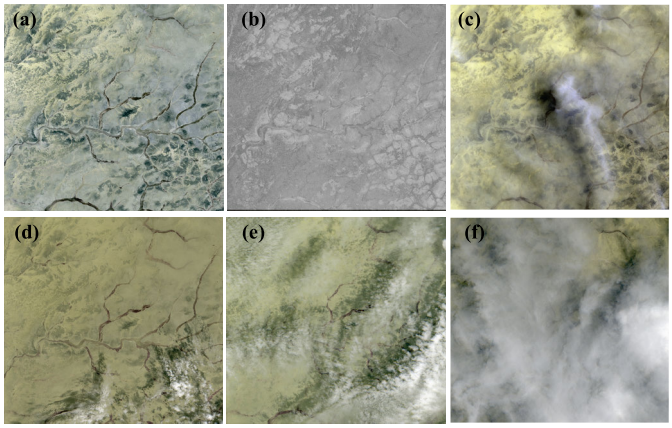


Fig. 4. Comparison of cloud states for different sub-datasets. (a) Cloud-free. (b) SAR. (c) TNC. (d) SPC. (e) LPC. (f) TKC.

B. Experimental Settings

1) Parameter Settings: The network is trained using the Adam optimizer. The learning rate is controlled using a cosine annealing strategy, with a default learning rate set of 1×10^{-4} and a minimum learning rate of 1×10^{-7}. During training, the batch size is set to 8, and the number of iterations is 300 epochs. The DG^2-TCR is implemented on Pytorch v1.8.0 + cu111, CUDNN v11.1, and the used GPU is GeForce RTX3090.

2) Evaluation Standards: Assume X and Y are the cloud-removed and reference ground truth images, respectively. H and W are the height and width, respectively. The scores of PSNR [23] and [58], RMSE, SSIM [59], SAM, NCC, and FID can be calculated to evaluate the cloud-removal image quality.

a) PSNR: One of the most commonly used metrics for evaluating image quality. PSNR is calculated based on the

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mse, defined in (14). Then, PSNR can be expressed by (15):

$$\text{mse} = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} \left[X(i, j) - Y(i, j)\right]^2 \quad (14)$$

$$\text{PSNR}(X, Y) = 10 \log_{10} \left[\frac{(2^n - 1)^2}{\text{mse}}\right] \quad (15)$$

where $2^n - 1$ represents the maximum value that a pixel in the image can take, and the pixels of a general image are 8-bit. The higher the PSNR, the closer the cloud-removed image is to the cloud-free reference image, which also indicates that the cloud-removal effect of the model is better.

b) RMSE: A commonly used indicator to measure the difference between a cloud-removed image and an actual cloud-free image, used to evaluate the degree of fit of the model on the given data. It can be calculated as follows:

$$\text{RMSE}(X, Y) = \sqrt{\text{mse}}. \quad (16)$$

c) SSIM: It is an essential indicator for evaluating the similarity between two images. SSIM comprehensively analyzes the similarity between two images by calculating the brightness, contrast, and image structure between the two images. The value of SSIM ranges from [0, 1]. The larger the value, the more similar the two images are. C_1 and C_2 are constants used to prevent division by zero.

$$\text{SSIM}(X, Y) = \frac{(2\mu_X\mu_Y + C_1)(2\sigma_{XY} + C_2)}{\left(\mu_X^2 + \mu_Y^2 + C_1\right)\left(\sigma_X^2 + \sigma_Y^2 + C_2\right)}. \quad (17)$$

d) NCC: It is an indicator that measures the similarity between two images, with values ranging from [−1, 1]. When used for cloud removal image quality assessment, it can effectively reflect the similarity of recovered texture details and other aspects of the reconstructed image. The higher the NCC, the better the recovery of detail information

$$\text{NCC} = \frac{1}{H \times W} \sum_{i,j} \frac{(X(i, j) - \mu_X)(Y(i, j) - \mu_Y)}{\sigma_X\sigma_Y}. \quad (18)$$

e) SAM: It measures the spectral similarity between the reconstructed and reference images by calculating the spectral vector angle between the two. The smaller the angle, the more similar the spectra

$$\text{SAM} = \cos^{-1}\left(\frac{Y^T X}{\|Y\|\|X\|}\right). \quad (19)$$

f) FID: The Fréchet inception distance (FID) is an effective metric for measuring the difference between generated images and real images in GAN

$$\text{FID}(X, Y) = \|\mu_X - \mu_Y\|_2^2 + Tr\left(\sigma_X + \sigma_Y - 2(\sigma_X\sigma_Y)^{1/2}\right). \quad (20)$$

In the above equations, μ_X and μ_Y, σ_X and σ_Y are the mean and variance of image X and Y, respectively. σ_{XY} is the covariance between X and Y.

3) Comparative Methods: To validate the performance of our method, we select five recently proposed deep learning-based algorithms for comparative analysis: NAFNet [34], SpA-GAN-CR [60], SAR-opt-Cgan [31], McGAN [61], USSRN-CR [49], cloud-attention GAN (CA-GAN) [62], and IDF-CR [63].

The NAFNet [34], SpA-GAN-CR [60], and IDF-CR [63] are based on ORSI for cloud removal tasks. While NAFNet is not inherently designed for remote sensing, it has shown outstanding performance in denoising tasks in computer vision. SpA-GAN-CR [60] utilizes local and global spatial attention mechanisms to identify and focus on cloud regions, thereby enhancing information recovery. IDF-CR [63] consists of a pixel space cloud removal module (Pixel-CR) and a latent space iterative noise diffusion network. McGAN [61] uses the near-infrared and visible light bands for cloud removal. SAR-opt-cGAN [31], USSRN-CR [49], and CA-GAN [62] use SAR images to guide the information reconstruction of the cloud-occluded area of the ORSI. SAR-opt-cGAN is the pioneering cGAN architecture that integrates SAR and optical multispectral images. USSRN-CR is based on a novel unified spatial-spectral residual network, which utilizes SAR images to accomplish cloud removal tasks. CA-GAN [62] is an end-to-end model that consists of a translation module, an attention module, a generator, and a discriminator. All the methods are trained on the two datasets for comparison analysis.

C. Comparative Analysis on SEN12MS-CR-TS Sub-Datasets

The processed datasets (TNC, SPC, LPC, TKC) are input for training. The cloud-free images are used as ground truth for loss computation. The ground truth images and their corresponding cloud images are geographically paired, with the acquisition time interval kept minimal to ensure reliability in performance evaluation. We randomly select three patches from different scene datasets for visual inspection. Fig. 5 depicts the comparative methods results on the four sub-datasets to validate the adaptive cloud removal capability.

In TNC, most methods can achieve good results in cloud removal. The SpA-GAN-CR, SAR-opt-cGAN, McGAN, and IDF-CR exhibit slightly weaker overall tonal style correction capabilities. However, they can accomplish the fundamental task of TNC removal.

In the SPC, the NAFNet, McGAN, USSRN-CR, CA-GAN, and our models successfully generate restored images with smooth textures and high-frequency details, showing relatively good cloud removal effectiveness. Overall, these methods demonstrate relatively good cloud removal effectiveness for small-scale fragmented clouds. Conversely, the SpA-GAN-CR and SAR-opt-cGAN gradually show defects such as incomplete removal of cloud blocks and unreasonable reconstruction results. They are not able to recover the missing information. The correlation between the cloud-removed and the cloud-free images of these two methods progressively widened the gap. In contrast, the proposed DG^2-TCR method effectively eliminates cloud blocks and stably restores texture details. As evident from Table III, the merits of the proposed method gradually come to the fore in SPC.

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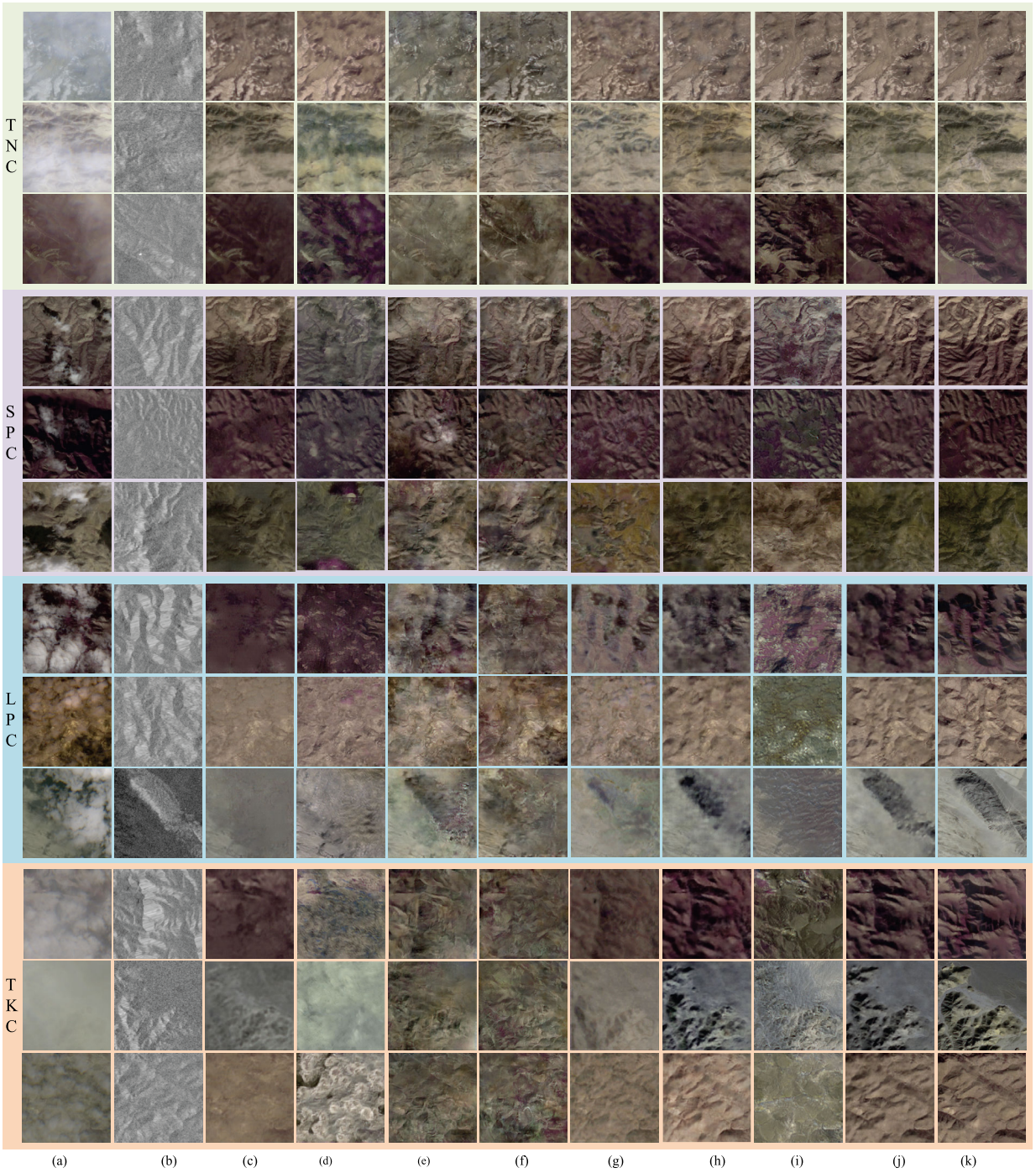


Fig. 5. Comparison results on TNC, SPC, LPC, and TKC. (a) Cloud. (b) SAR. (c) NAFNet. (d) SpA-GAN-CR. (e) SAR-opt-cGAN. (f) McGAN. (g) USSRN-CR. (h) CAGAN. (i) IDF-CR. (j) Ours. (k) Cloud-free.

In the LPC, as the cloud coverage area expands, the challenge of cloud removal intensifies. NAFNet removes TKCs and provides smooth filling but fails to restore the intricate details beneath the clouds accurately. SpA-GAN-SAR model succeeds in cloud removal but struggles with restoring texture details. SAR-opt-cGAN, McGAN, and IDF-CR models exhibit limited cloud removal capabilities, and their reliance on GAN leads to unstable performance, rendering their de-clouding results erratic and ineffective. Furthermore, neither model is sensitive to color space information and cannot restore the hue to be consistent with the target image based on cloud removal. The USSRN-CR and CAGAN model achieves notably better cloud removal overall, yet its texture detail restoration remains limited. DG^2-TCR continues to demonstrate a robust cloud removal performance. Though the resolution of objects beneath large-scale clouds diminishes slightly during texture restoration, the proposed method surpasses its competitors in terms of removal effectiveness and tone matching.

TKC removal results are shown in the TKC dataset (Fig. 5). In such scenarios, NAFNet, SpA-GAN-CR, and IDF-CR,

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TABLE III
QUANTITATIVE EVALUATION RESULTS

Metric	Method	TNC	SPC	LPC	TKC	Avg	SMILE-CR
PSNR ↑	NAFNet	29.694	30.829	30.528	29.341	30.098	38.378
	SpA-GAN-CR	28.719	29.356	29.303	28.209	28.896	29.989
	SAR-opt-cGAN	28.158	28.157	28.105	28.059	28.120	28.613
	McGAN	28.143	28.291	28.200	28.040	28.168	28.595
	USSRN-CR	29.195	29.898	29.556	28.846	29.374	37.964
	CAGAN	28.812	30.268	29.705	28.796	29.395	35.194
	IDF-CR	28.935	29.513	28.975	28.438	28.965	29.933
	ours	**29.880**	**31.499**	**30.833**	**29.436**	**30.412**	**38.596**
RMSE ↓	NAFNet	0.0733	0.0639	0.0847	0.0921	0.0785	0.0379
	SpA-GAN-CR	0.0714	0.0686	0.0795	0.0869	0.0766	0.0455
	SAR-opt-cGAN	0.0841	0.1052	0.0878	0.0714	0.0871	0.0450
	McGAN	0.0722	0.0982	0.1001	0.0856	0.0890	0.0413
	USSRN-CR	0.0738	0.0689	0.1087	0.0779	0.0823	0.0379
	CAGAN	0.0910	0.0705	0.0815	0.0864	0.0824	0.0420
	IDF-CR	0.0684	0.0648	0.0801	0.0878	0.0753	0.0419
	ours	**0.0680**	**0.0596**	**0.0790**	**0.0729**	**0.0699**	**0.0306**
SSIM ↑	NAFNet	0.7448	0.7573	0.7072	0.6255	0.7087	0.6801
	SpA-GAN-CR	0.6444	0.6804	0.5855	0.6340	0.6361	0.5609
	SAR-opt-cGAN	0.6503	0.6298	0.5664	0.4748	0.5803	0.5416
	McGAN	0.6403	0.6347	0.5608	0.4097	0.5614	0.5555
	USSRN-CR	0.7064	0.7211	0.6463	0.5837	0.6644	0.6791
	CAGAN	0.6389	0.7746	0.7108	0.6414	0.6914	0.7177
	IDF-CR	0.6748	0.6934	0.6749	0.4954	0.6346	0.6646
	ours	**0.7934**	**0.8127**	**0.7460**	**0.6713**	**0.7559**	**0.7324**
SAM ↓	NAFNet	1.3091	1.5103	1.3580	1.1338	1.3278	0.5281
	SpA-GAN-CR	0.8588	1.080	1.7855	0.9732	1.1744	0.7008
	SAR-opt-cGAN	**0.5183**	**0.6796**	**0.5636**	**0.8504**	**0.6530**	**0.3475**
	McGAN	0.6835	0.8794	0.8968	0.9025	0.8406	0.4424
	USSRN-CR	0.9461	1.1070	1.2305	1.3308	1.1536	0.4638
	CAGAN	1.5277	1.4353	1.9695	1.5255	1.6145	0.7202
	IDF-CR	0.8840	1.3947	0.8311	1.2139	1.0810	0.8550
	ours	0.8923	1.1391	1.1580	1.1107	1.0750	0.4560
NCC ↑	NAFNet	0.6906	0.7539	0.6409	0.4694	0.6387	0.556
	SpA-GAN-CR	0.4973	0.5948	0.4366	0.1930	0.4304	0.4233
	SAR-opt-cGAN	0.5150	0.4634	0.4231	0.2522	0.4134	0.3705
	McGAN	0.5023	0.5357	0.4134	0.2206	0.4180	0.3915
	USSRN-CR	0.6031	0.6846	0.5723	0.4175	0.5694	0.5548
	CAGAN	0.5451	0.7477	0.6407	0.5357	0.6173	0.6533
	IDF-CR	0.5528	0.6280	0.2878	0.2755	0.4360	0.6253
	ours	**0.7330**	**0.7928**	**0.7084**	**0.5996**	**0.7084**	**0.667**
FID ↓	NAFNet	194.68	152.76	209.86	271.92	207.31	172.61
	SpA-GAN-CR	246.22	189.62	249.82	325.44	252.78	262.33
	SAR-opt-cGAN	213.50	215.16	242.56	290.86	240.52	254.87
	McGAN	213.62	195.26	238.75	302.33	237.49	252.08
	USSRN-CR	220.45	174.35	233.23	296.78	231.20	173.47
	CAGAN	265.96	154.10	212.86	266.96	224.97	160.59
	IDF-CR	214.45	185.01	274.35	284.73	239.64	195.96
	ours	**167.50**	**136.30**	**187.41**	**249.05**	**185.06**	**158.02**

↑ :The higher score indicates the better effect; ↓ :The lower score indicates the better effect; The best value of each indicator is bolded.

which rely only on RGB data, failed to restore cloud details. Although McGAN, utilizing near-infrared and RGB bands, recovered some texture details, the results are insignificant. The SAR-opt-cGAN model incorporated SAR data, allowing it to retrieve some ground object information when the image is not completely occluded. However, its extreme instability resulted in inconsistent cloud removal performance. The USSRN-CR exhibited better in cloud removal but suffered from drawbacks like subpar tone migration and blurred texture detail restoration due to the residual structure employed by USSRN-CR. CAGAN also shows a similar recovery with the ground truth. However, the details are not as well recovered as our methods. Conversely, DG2-TCR demonstrated exceptional cloud-removal capabilities, efficiently eliminating clouds while matching the image tone to the target. Nevertheless, the resolution of DG2-TCR de-clouded images was lower than that of cloud-free images. As Fig. 6 illustrates, as the cloud coverage area increases, the correlation coefficients of all models decline, indicating a more significant deviation from the diagonal. Nevertheless, the cloud-removal effectiveness of our proposed method remains consistent, indicating its superior missing information reconstruction performance in high-cloud coverage scenarios.

Fig. 6 shows the correlation between the cloud-removal images and the ground truth images. In contrast, the proposed DG^2-TCR method effectively eliminates cloud blocks and stably restores texture details. The correlation of all methods in TNC is better than in other scenes. The SpA-GAN-CR, SAR-opt-cGAN, and IDF-CR gradually show defects such as incomplete removal of cloud blocks and unreasonable reconstruction results. Other methods that introduce SAR show better recovery. Compared to TKC scenarios involving SPC, LPC, and TKC, the correlation between the proposed method of cloud removal results and the truth results is slightly reduced. Despite this slight reduction in correlation, the proposed method of cloud-removal performance remains consistently high. The proposed method performs better on the TNC, SPC, LPC, and TKC datasets compared to several other comparison methods.

D. Comparative Analysis of SIMLE-CR

Fig. 7 presents the results of different methods on the SMILE-CR dataset. This dataset is relatively small, with only 1000 training images. Methods that rely solely on RGB data, such as NAFNet and SpA-GAN-CR, almost cannot remove the cloud. Although McGAN recovers some texture details using near-infrared and RGB bands, the results remain poor. The SAR-opt-cPAN model combines SAR data, enabling it to retrieve some ground object information when the image is not fully occluded. However, its extreme instability leads to inconsistent cloud-removal performance. USSRN-CR should theoretically perform better in cloud removal, but results show that its cloud-removal effect on this dataset is significantly worse than our approach. Although CAGAN eliminates some clouds, the repaired result exhibits noticeable spectral color differences compared to the ground truth image. Similarly, the IDF-CR also removes some clouds on the image, with the restored result showing a similar spectral color difference to the ground truth. However, its recovery of details under the cloud is notably poor, and restoring feature details is not ideal. In contrast, our method, DG^2-TCR, demonstrates excellent cloud removal capability in this dataset as well, effectively eliminating the clouds while matching the ground truth. However, the resolution of the cloud-removed images from DG^2-TCR is lower than that of the ground truth images. As shown in Fig. 8, with the increase in cloud coverage, the correlation coefficients of all models decrease, indicating a greater deviation from the diagonal. However, the cloud removal effect of our proposed method remains consistent, indicating its superior performance in missing information reconstruction in high cloud coverage scenarios.

E. Quantitative Comparative Analysis

Table III shows the quantitative results of four sub-datasets in SEN12MS-CR-TS and SIMLE-CR. Our method performs best on PSNR, RMSE, SIM, FID, and NCC. SAR-opt-cGAN

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Fig. 6. Scatter diagrams between the original and reconstructed pixels in Fig. 5. *R* represents the correlation between the original image and the reconstructed image. (a) NAFNet. (b) SpA-GAN-CR. (c) SAR-opt-cGAN. (d) McGAN. (e) USSRN-CR. (f) CAGAN. (g) IDF-CR. (h) Ours.

got the best SAM, and our method is the third best. The total results show our method is the best and most stable. No matter which kinds of clouds cover, our approach can achieve the best quality, proving that its self-adaption ability to remove different clouds. However, the other methods are not stable. NAFNet emerges as a clear winner in TNC and SPC scenarios. However, as the cloud occlusion intensity and coverage grow, the model's approach of smoothly filling in missing information becomes ineffective. Models such as SpA-GAN-CR, SAR-opt-cGAN, McGAN, and CAGAN all leverage generative adversarial network (GAN) structures, but the inherent instability of this architecture leads to inconsistent cloud-removal performance. While SAR-opt-cGAN stands apart by incorporating real-world terrain prior

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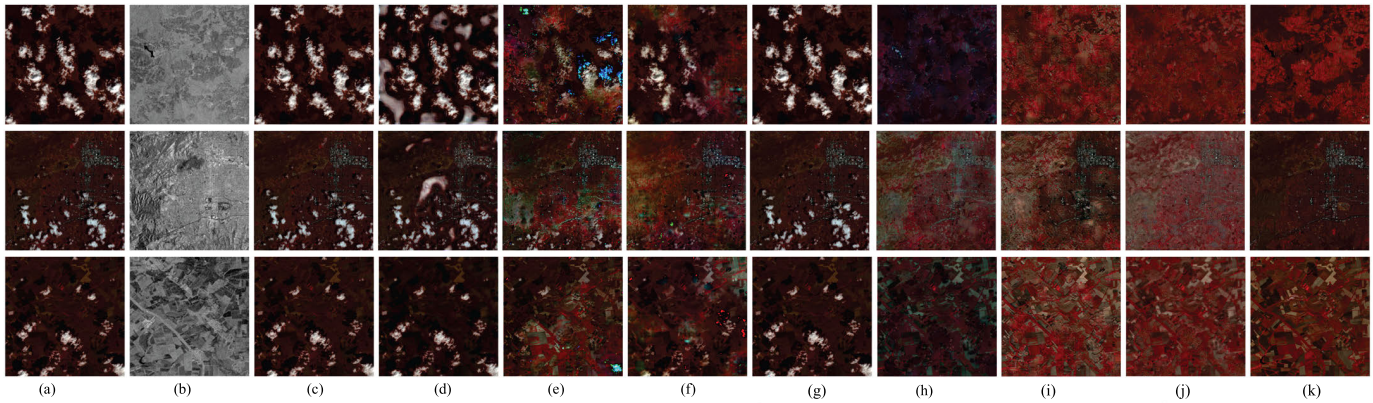


Fig. 7. Experimental result on SMILE-CR. (a) Cloud. (b) SAR. (c) NAFNet. (d) SpA-GAN-CR. (e) SAR-opt-Cgan. (f) McGAN. (g) USSRN-CR. (h) CAGAN. (i) IDF-CR. (j) Ours. (k) Cloud-free.

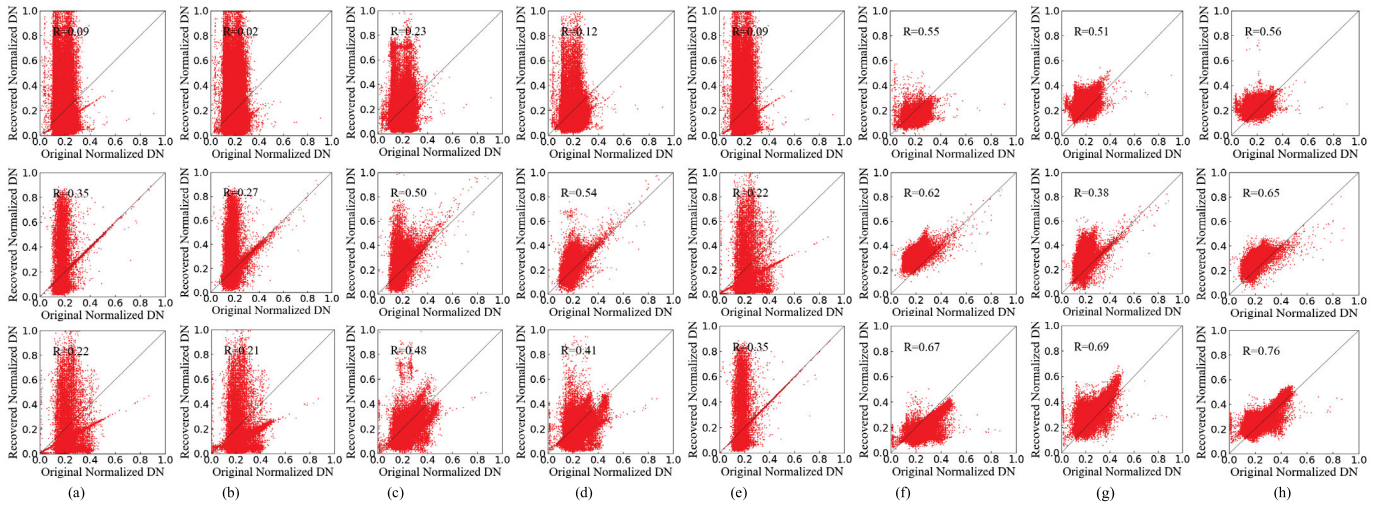


Fig. 8. Scatter diagrams between the original and reconstructed pixels in Fig. 7. *R* represents the correlation between the original image and the reconstructed image. (a) NAFNet. (b) SpA-GAN-CR. (c) SAR-opt-cGAN. (d) McGAN. (e) USSRN-CR. (f) CAGAN. (g) IDF-CR. (h) Ours.

knowledge from SAR data, enabling it to recover details to a certain extent, its effectiveness is still limited. The USSRN-CR model, based on a residual network structure, can effectively remove clouds from the remote sensing images. However, the recovered detailed information lacks significant global context, and the resolution falls short of meeting practical requirements. CAGAN also performed well, which was less than our results. Our proposed method achieves excellent cloud removal effects across TNC, SPC, LPC, and TKC scenarios. The methods located in the second place in each evaluation index are different. Our method boasts an average PSNR of 1.0384 dB higher and an SSIM of 0.0915 dB higher than the USSRN-CR model. The NCC and FID show that our methods achieved the best information recovery in detail. Thus, our method can remove clouds adaptively in different cloud scenarios.

V. Discussion

A. Ablation Studies

DG^2-TCR incorporates two core improvements, DFG and SSIRB. We execute ablation experiments on various datasets and analyze their effectiveness, thereby evaluating the importance of these modules.

1) Ablation Study of the DFG: DFG is a crucial component in encapsulating the global feature information of SAR images within the network, guiding the refinement of global texture features in the primary FR layer. We construct different network structures based on the inclusion or exclusion of DFG in the DG^2-TCR framework and quantitatively analyze the cloud removal effects of these structures across diverse scenarios to assess their performance. As shown in Fig. 9 and Table IV, the network with DFG is superior to the network without this structure in terms of PSNR, RMSE, SSIM, SAM, NCC, and FID. The experiments demonstrate that DFG effectively enhances the cloud-removal capability.

2) Ablation Study of the SSIRB: SSIRB is designed to extract contextual features from spatial and channel dimensions, enabling the recovery of ground information across multiple scales. The backbone of our method is constructed using SSIRB as the fundamental block. DAGC and CAB are the core components of SSIRB. To evaluate their impact, we replace DAGC and CAB with 3×3 convolutional layers and present the qualitative results in Fig. 10. Here, a random patch is selected from the four sub-datasets to compare the intermediate cloud removal effectiveness among

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TABLE IV
ABLATION RESULTS OF DFG

Data	DFG	PSNR ↑	RMSE ↓	SSIM ↑	SAM ↓	NCC ↑	FID ↓
TNC	no	29.1321	0.0764	0.755	0.822	0.6736	197.2603
	Δ add	0.7482	-0.008	0.0384	0.0703	0.0594	-29.7546
SPC	no	30.2101	0.0792	0.7801	1.1259	0.756	152.3909
	Δ add	1.2897	-0.0196	0.0326	0.0132	0.0368	-16.0877
LPC	no	29.6158	0.0976	0.7182	1.2418	0.6609	198.9823
	Δ add	1.2181	-0.0178	0.0278	-0.0838	0.0475	-11.5671
TKC	no	28.8953	0.1015	0.6344	1.1576	0.5282	242.7971
	Δ add	0.5411	-0.0286	0.0369	-0.0469	0.0714	6.2557
AVG	no	29.4633	0.0887	0.7219	1.0868	0.6547	197.8576
	Δ add	**0.9493**	**-0.0185**	**0.034**	**-0.0118**	**0.0537**	**-12.7884**

(Δ add) represents the difference between adding DFG and without DFG.

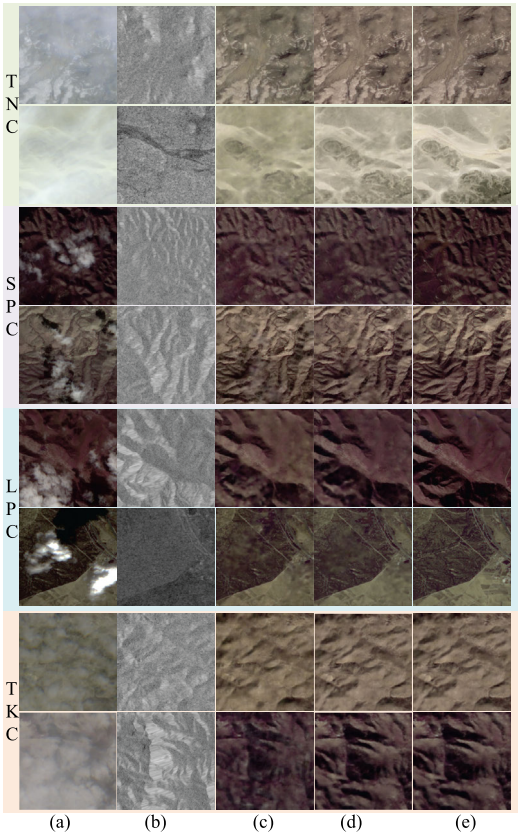


Fig. 9. Impact of DFG. (a) Cloud image. (b) SAR. (c) No DFG. (d) Add DFG. (e) Cloud-free.

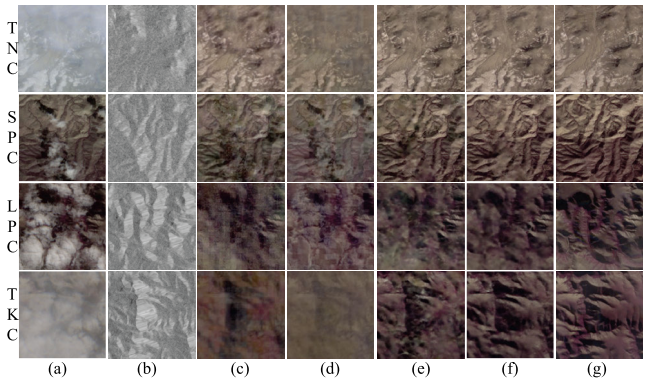


Fig. 10. Impact of different SSIRB structures. (a) Cloud image. (b) SAR. (c) Baseline. (d) Add CAB. (e) Add DAGC. (f) Ours. (g) Cloud-free.

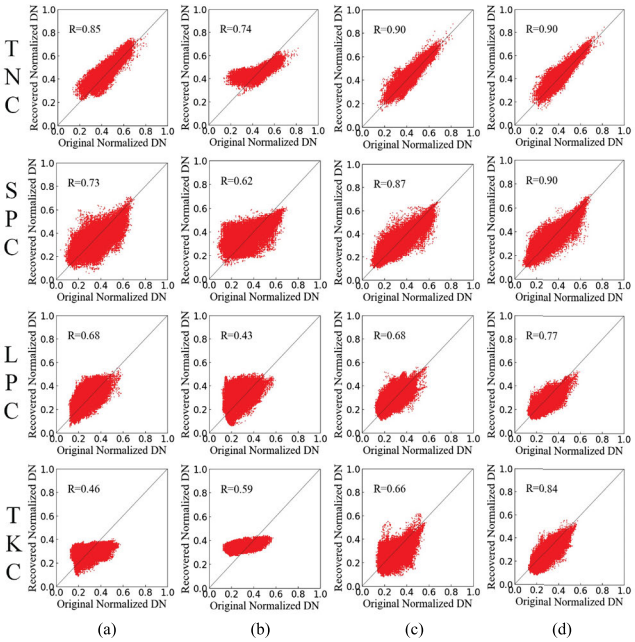


Fig. 11. Scatter diagrams between the original and reconstructed pixels in Fig. 10. (a) CNN. (b) Add CAB. (c) Add DAGC. (d) Ours. R represents the correlation between the original image and the reconstructed image.

various structures. Table IV quantitatively evaluates the average indexes of four sub-datasets among different networks. To gain deeper insights into the network performance influenced by CAB and DAGC, Fig. 11 displays the scatter plots and correlation coefficients of the cloud-removal results.

When CAB is introduced into SSIRB and DAGC is replaced by 3×3 convolutional layers, the model shows defects of information loss in cloud-occluded areas and reduced resolution of information in cloud-free regions of both thin and TKC removal tasks. Fig. 11 and Table V show that when CAB is introduced alone, the three evaluation indicators of PSNR, RMSE, SSIM, SAM, and NCC are all reduced, and the correlation between the cloud-removal results and the truth value is also reduced. When only DAGC is introduced into SSIRB, the model can achieve good results in TNC and SPC scenes. However, as the information content of cloud-free regions decreases, this model structure's missing information reconstruction capability gradually diminishes, exhibiting channel color differences. The quantitative evaluation results show that when only DAGC is introduced into the structure, the model cloud removal results are improved by 0.164 and 0.055 dB in PSNR and SSIM scores, respectively. The effect of improvement is not apparent. When DAGC and CAB are introduced into SSIRB, the model can achieve a better cloud-removal effect regardless of thin, broken, or TKC scenes. It can be seen from Table V that after introducing DAGC and CAB in SSIRB, the network cloud-removal results capability is improved by 0.895, 0.080, and 0.167 dB in PSNR, SSIM, and NCC scores, respectively. The RMSE, SAM, and FID scores decreased by 0.015, −0.063, and −52.127 dB. The results prove that combining DAGC and CAB effectively enhances the model's ability to reconstruct information in TKC scenarios.

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TABLE V
AVERAGE INDEX OF DIFFERENT SSIRB STRUCTURES

Method	Base Line (CNN)	CAB	DAGC	PSNR ↑	RMSE ↓	SSIM ↑	SAM ↓	NCC ↑	FID ↓
(i)	√			29.518	0.085	0.676	1.138	0.541	237.196
(ii)	√	√		28.702	0.106	0.646	**1.069**	0.453	260.879
(iii)	√		√	29.682	0.088	0.732	1.091	0.644	194.484
(iv)	√	√	√	**30.417**	**0.070**	**0.756**	1.075	**0.708**	**185.069**

TABLE VI
QUANTITATIVE EVALUATION OF CLOUD-REMOVAL EFFECT BASED ON DIFFERENT SOURCES OF GLOBAL FEATURES

Feature	PSNR ↑	RMSE ↓	SSIM ↑	SAM ↓	NCC ↑	FID ↓
ORSI	29.206	0.084	0.575	1.2737	0.535	249.74
SAR	**29.436**	**0.073**	**0.671**	**1.2107**	**0.600**	**249.05**
SAR+ORSI	29.257	0.078	0.580	1.2363	0.538	249.55

TABLE VII
COMPARISON OF THE INFERENCE TIME BY DIFFERENT METHODS FOR INFERRING A SINGLE BATCH

Methods	Time/s	Methods	Time/s
NAFNet	0.3488	USSRN-CR	0.9032
SpA-GAN-CR	0.4114	CAGAN	1.840
SAR-opt-cGAN	1.3224	IDF-CR	1.889
McGAN	0.9664	our	1.4754

TABLE VIII
QUANTITATIVE INDICATORS OF THE MIGRATION EFFECT OF DG^2-TCR IN DIFFERENT CLOUD SCENARIOS

Source	Target	ΔPSNR	ΔRMSE	ΔSSIM	ΔSAM	ΔNCC	ΔFID
TNC	SPC	-1.3943	0.0043	-0.1054	0.109	-0.126	6.484
	LPC	-1.6370	0.0301	-0.1894	0.114	-0.296	67.698
	TKC	-1.5459	0.0144	-0.226	-0.059	-0.369	101.346
SPC	TNC	-3.3504	0.0561	-0.1448	-0.391	-0.242	95.369
	LPC	-2.7645	0.0440	-0.154	0.3003	-0.233	84.607
	TKC	-3.2644	0.0409	-0.231	-0.005	-0.411	139.889
LPC	TNC	-2.6992	0.0403	-0.123	-0.526	-0.169	52.798
	SPC	-2.0211	0.0127	-0.051	-0.089	-0.067	2.759
	TKC	-2.5429	0.0378	-0.167	-0.064	-0.313	78.089
TKC	TNC	-0.7606	0.0101	-0.0004	-0.364	0.007	-28.682
	SPC	-0.6773	0.0335	-0.0014	0.052	0.048	-70.034
	LPC	-0.6336	0.0841	-0.013	0.169	-0.042	-26.690

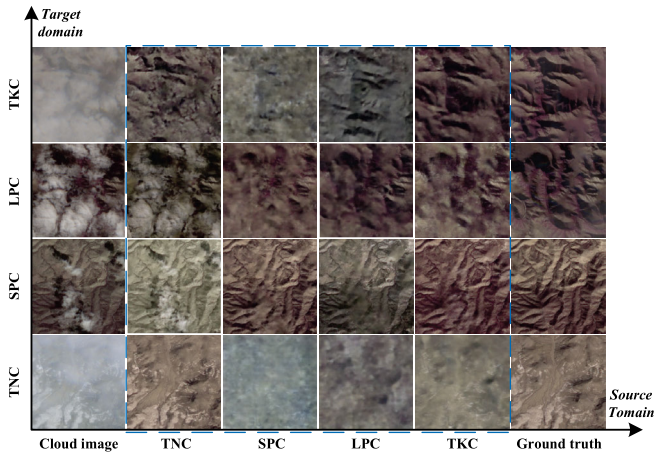


Fig. 12. Migration effect of DG^2-TCR in different cloud scenarios. The horizontal and vertical axis are the training and target dataset, respectively.

B. Ablation Study of the SAR Data

In the global texture feature encapsulation module, global features originate from the input image. Three distinct encapsulation blocks are tested, including leveraging combinations of ORSI, SAR, and their fusion. Our approach directly harnesses SAR images for FE. Table VI presents test results on the TKC dataset, revealing that encapsulating global features solely from SAR images enhances the network cloud-removal capability. This is attributable to SAR's ability to provide practical ground object texture features, even when TKCs completely obscure the ground. Conversely, ORSI loses all information under such conditions, and fusing SAR with optical images introduces substantial interference, hampering the provision of genuine global features for texture correction, ultimately resulting in inferior cloud-removal outcomes.

C. Inference Time

Table VII compares the inference results of different methods. Since the baseline networks of the comparison methods include different types of networks such as GANs and CNNs, directly comparing parameters, floating-point operations (FLOPs), and other metrics is meaningless. Therefore, the advantages of the proposed method in terms of complexity and time efficiency are validated by comparing the time consumed for inference with the same batch size under the same device and hyperparameter settings. The results are shown in the table. The time consumed by the proposed method is similar to that of several other compared methods and even lower than that of methods such as IDF-CR and CAGAN. After a comprehensive comparison and analysis of complexity and inference time, although the proposed method shows average performance in terms of inference efficiency, it achieves better cloud-removal results.

D. Robustness Analysis of DG^2-TCR

DG^2-TCR training in specific cloud scenarios as a source domain can be migrated to other diverse target domains. As shown in Fig. 12, the migration efficacy of DG2-TCR is tested by cross-domain scenarios. The precise quantitative analysis of its performance has been compiled and summarized in Table VIII. Additionally, it presents the shift in the test score observed on the target domain dataset, derived from the benchmark result obtained on the source dataset.

The overall cloud removal performance is suboptimal when the model train on TNC training is directly transferred to other scenarios (SPC, LPC, TKC). Specifically, in TKC scenes with extremely low cloud transparency, effectively accomplishing cloud-removal tasks poses a significant challenge. Conversely, when the model trained for TKC is directly applied to TNCs, dense clouds, and large-scale broken cloud occlusion scenes, although the cloud-removal effect is reduced, it can still remove cloud-removal ability. From Table VIII, it can be observed that when TKC is used as the training dataset, the decrease in PSNR and SSIM scores is the smallest compared to the other experimental groups. Therefore, the generalization

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performance of the model trained on TKC data effectively removes clouds of varying shapes and coverage ranges.

VI. Conclusion

This article presents the novel cloud-removal network, DG^2-TCR, which leverages SAR to recover missing information through DFG. DG^2-TCR harnesses SAR data as a priori knowledge base to address the challenge of cloud removal from ORSI effectively. DFG serves two primary purposes. On the one hand, it separates and extracts cloud and noncloud features locally, effectively recovering texture features in cloudy areas. On the other hand, it encapsulates SAR data into global features to guide the feature compensation of texture details. In addition, this article designs a combination of DAGC and CAB to build an SSIRB module, which can implicitly distinguish cloud and noncloud pixels and reconstruct missing features in the cloud area. To validate the cloud-removal effectiveness of the proposed method under varying cloud coverage conditions, the SEN12MS-CR-TS dataset was divided into four sub-datasets based on factors such as cloud morphology and coverage ratio. In addition, SIMLE-CR is also tested. Our method demonstrates a significantly superior cloud-removal effect compared to other comparative methods across the datasets, which got the best values in the PSNR, RMSE, SSIM, NCC, and FID indicators. Furthermore, through generalization experiments, it has been demonstrated that the stability and adaptability of the proposed method are better than other methods. Test results exhibit strong cloud removal performance as the extent of cloud occlusion increases and TKCs become more prevalent.

Despite our method achieving promising qualitative and quantitative results, certain limitations still warrant consideration. For future research, we intend to delve into utilizing diverse data sources for optical remote sensing image cloud-removal tasks to devise even more effective models for tackling the challenge of TKC removal.

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